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healthdirect Australia is a free service where you can talk to a nurse or doctor who can help you know what to do.

Diabetes

9-minute read

Listen

Key facts

- Diabetes refers to a group of diseases that can cause high levels of glucose (a type of sugar) in the blood.
- Diabetes develops when your pancreas can't produce enough of the hormone insulin or your body becomes resistant to it.
- Symptoms of diabetes include feeling tired, hungry or excessively thirsty, and passing more urine than usual.
- Common diabetic conditions include type 1 diabetes, type 2 diabetes, gestational diabetes and pre-diabetes.
- You can manage diabetes by taking medicines to manage your blood glucose levels, adopting a healthy diet and being physically active.

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What is diabetes?

Diabetes is a health condition in which your body has difficulty converting glucose (a type of sugar) into energy. This leads to high levels of sugar in the blood ([hyperglycaemia](#)).

Your blood glucose levels are normally controlled by a [hormone](#) called insulin, which converts glucose into energy. Diabetes occurs when your [pancreas](#) can't produce insulin or when your body can't make use of the insulin because it's grown resistant to it.

There are several types of diabetes:

- [Type 1 diabetes](#), occurs when a person's own [immune system](#) breaks down the cells in the pancreas that produce insulin.
- [Type 2 diabetes](#) occurs when the body's cells do not respond effectively to insulin or lose the ability to produce enough insulin in the pancreas. This causes glucose to stay in the blood, leading to a higher-than-normal level of glucose in the body.
- [Gestational diabetes](#) occurs when a woman experiences high blood glucose levels during [pregnancy](#). It usually goes away after the baby is born. Women who experience gestational diabetes have a higher risk of developing type 2 diabetes in the future.
- [Pre-diabetes](#) is where blood glucose levels are higher than usual, but not high enough for a diagnosis of diabetes.

What are the symptoms of diabetes?

Common symptoms of diabetes include:

- excessive thirst or hunger
- passing more urine than usual
- feeling tired and lethargic
- unexplained weight loss (for type 1), or gradual weight gain (for type 2)

- having cuts that heal slowly
- itchy skin or skin infections
- blurred vision

Type 1 diabetes is usually spotted quickly, since symptoms can appear suddenly. Most people with type 1 diabetes are diagnosed before the age of 19.

Many people with type 2 diabetes, however, don't have symptoms at all or have signs that go unnoticed for a long time.



CHECK YOUR SYMPTOMS — Use the diabetes [Symptom Checker](#) and find out if you need to seek medical help.

When should I see my doctor?

Diabetes is serious but can be managed well with a combination of [medicines](#) and lifestyle changes. Early diagnosis and treatment can also help to reduce the risk of more serious complications. Speak to your doctor if you are experiencing symptoms of diabetes or have concerns about your risk of diabetes.



ARE YOU AT RISK? — Are you at risk of type 2 diabetes? Use the [Risk Checker](#) to find out.

What causes diabetes?

There are different possible causes of diabetes according to type.

Type 1 diabetes

Type 1 diabetes is an [auto-immune condition](#): your body's immune system attacks the cells in your pancreas that produce insulin. The exact cause of this reaction is still unknown, but diet and lifestyle are not factors that determine who gets type 1 diabetes. Research suggests that both genetics and the environment may play a part. While having a [family history](#) increases your risk, most people with type 1 diabetes have no family history of the condition.

Type 2 diabetes

Type 2 diabetes develops gradually over years as your body's insulin becomes less effective at managing your blood glucose levels. As a result, your pancreas produces more and more insulin, and eventually the insulin producing cells wear out and become ineffective. Type 2 diabetes is a combination of low insulin and ineffective insulin.

The risk of getting type 2 diabetes increases with certain lifestyle factors:

- a family history of diabetes
- low levels of [physical activity](#).
- poor [diet](#)
- being [obese](#) or overweight, especially [around the waist](#)



NEED TO LOSE WEIGHT? — Use the [BMI Calculator](#) to find out if your weight and waist size are in a healthy range.

Gestational diabetes

During pregnancy, certain hormones that provide nutrition for a growing baby reduce the effectiveness of the mother's own insulin. If the pancreas is unable to produce enough insulin in response, blood glucose levels rise and gestational diabetes develops.

How is diabetes diagnosed?

If your doctor suspects you have diabetes, you will be asked to have a [blood test](#) to check your glucose levels.

This may include:

- **Fasting glucose test** — testing your glucose levels after fasting for 8 hours.
- **Oral glucose tolerance test (OGTT)** — this involves drinking a sugary drink (after fasting), then testing your glucose levels 1 to 2 hours later to assess the effect.
- **Random blood glucose test** — a blood test taken without fasting.
- **HbA1c test** — this test doesn't require fasting as it doesn't test glucose levels directly, but looks for a by-product [red blood cells](#) that shows how your body manages glucose over time.



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If you're pregnant, your doctor will screen for gestational diabetes as part of standard [antenatal testing](#).

How is diabetes treated and managed?

Although there is currently no cure, diabetes can be managed with lifestyle changes and medication.

Type 1 diabetes

If you have type 1 diabetes, you will need [insulin replacement](#), through injections or an [insulin pump](#). This helps control your body's blood glucose levels. While there's nothing you can do to prevent type 1 diabetes, your lifestyle choices after your diagnosis can reduce the risk of developing more serious complications such as [kidney disease](#), [eye damage](#) and [foot problems](#).

Type 2 diabetes

Type 2 diabetes can sometimes be initially managed by lifestyle changes, such as following a healthy diet and being physically active. Eventually, you may need to take medicines to manage glucose levels. Your doctor may prescribe tablets or injectable medicines (insulin or others) to help keep your blood glucose in the [target range](#).

Gestational diabetes

While every woman's experience is different, gestational diabetes can usually be managed by keeping blood glucose levels within the range for a healthy pregnancy. This involves a [healthy eating plan](#) and regular physical activity. Medication may be needed if adequate glucose control can't be achieved with lifestyle changes alone.

Blood glucose monitoring

Crucial to diabetes management is regular [blood glucose monitoring](#). This helps you to understand the relationship between blood glucose, food, exercise and insulin in your body. It also helps you keep track of potential treatment side effects, such as [hypoglycaemia](#) (a sudden drop in blood sugar).

Watch this video to learn how to manage your diabetes. The video is available in [multiple languages](#).



Managing my diabetes

11:53

Taking steps to manage your diabetes can help you stay healthy and happy.

Video provided by [Diabetes Victoria](#).

Can diabetes be prevented?

While type 1 diabetes can't be prevented, there is clear evidence that maintaining a healthy weight, exercising regularly and following a healthy eating plan can delay or even prevent type 2 diabetes.

Healthy habits that can help prevent type 2 diabetes include:



- [healthy eating](#)
- maintaining a [healthy weight](#), or [losing weight](#) if you are overweight or obese
- regular physical activity
- quitting smoking
- reducing your [alcohol intake](#) (fewer than 4 standard drinks a day)

See [your doctor](#) for regular health checks, or join a prevention program in your state to connect with someone who can help:

- NSW — Sign up to the [Get Healthy Service](#). They offer a program specifically designed for people at high risk of developing type 2 diabetes, as well as providing information on improving your general health.
- NT — Visit [Healthy Living NT](#) for community and group diabetes education sessions.
- Queensland — Visit [My Health for Life](#). This is a free, six-month program where you work with a health coach to achieve your health goals.
- Tasmania — Sign up to the [COACH Program](#). This is an evidence-based, award-winning coaching prevention program for people with chronic disease or at high risk of chronic disease.
- Victoria — [Visit The Life! Program](#) is a free Victorian lifestyle-modification program that helps you reduce your risk of type 2 diabetes and cardiovascular disease.
- WA — Learn about the [Let's Prevent](#) is a free online health program that will help you set a personal plan to get your body healthy and prevent type 2 diabetes, heart disease and stroke.

Are there complications of diabetes?

Diabetes is a complex disease that can affect your whole body. Some complications of diabetes include:

- **Heart disease and stroke:** people with diabetes are up to 4 times more likely to suffer a [heart attack](#) or [stroke](#).
- **High blood pressure:** people with diabetes may develop [high blood pressure \(hypertension\)](#) due to changes to their body chemistry.
- **Foot problems:** having diabetes increases your risk of foot and [leg ulcers](#) and [amputations](#) — see a [podiatrist](#) regularly.
- **Vision problems:** diabetes is the leading cause of preventable [blindness](#) in Australia — you can register for eye check reminders at [KeepSight Australia](#).
- **Kidney disease:** [kidney failure](#) is 3 times more common in people with diabetes.

Diabetes can have a big impact on mental health as well — more than 3 in 10 people with diabetes experience [depression](#), [anxiety](#) and distress.

There are many ways you can reduce your risk of complications. Talk to your doctor or a diabetes educator about how to look after yourself to minimise — or even prevent — diabetes complications.

Resources and support

For more information and support, try these resources:

- Diabetes Australia's website has information about [the different types of diabetes](#) and practical advice for [living with diabetes](#).
- Call the [National Diabetes Services Scheme \(NDSS\)](#) helpline on 1800 637 700 to access a range of subsidised products to help you manage your diabetes.

Aboriginal and Torres Strait Islander people

For culturally-sensitive resources, try:

- Diabetes Australia's page on [Diabetes in Aboriginal and Torres Strait Islander communities](#) (includes a video)
- Healthy Living NT's diabetes factsheet on [keeping your feet healthy](#).
- Healthy Living NT's factsheet on [nutrition for diabetes in pregnancy](#).

Other languages

Do you prefer reading in languages other than English? These websites offer translated information about diabetes:

- [NDSS Multicultural Portal](#)
- [Multicultural Health Communication Service NSW](#)
- [Health Translations Victoria](#)

Apps and tools

You might find these apps and tools helpful:

- [healthdirect's Risk Checker](#) (diabetes, heart disease and kidney disease)
- [Diabetes Australia's risk calculator](#)
- [Diabetes Australia's mobile app](#)

Sources:

[Department of Health](#) (*What we're doing about diabetes*), [Juvenile Diabetes Research Foundation](#) (*What is type 1 diabetes*), [Lab Tests Online](#) (*HbA1c*), [The Royal Australian College of General Practitioners](#) (*Management of type 2 diabetes: A handbook for general practice*), [Diabetes Australia](#) (*Home*)



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Learn more here about the [development and quality assurance of healthdirect content](#).

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